

ALGORITHM

- STEP 1: Set up a message "Enter a string: "
- STEP 2: Set up a message "The given string is not a Palindrome"
- STEP 3: Set up a message "The given string is a Palindrome"
- STEP 4: Create a buffer to hold input string
- STEP 5: Print the message "Enter a string:"
- STEP 6: Read a string input from the user
- STEP 7: Set pointers to the start and end of the input string
- STEP 8: Initialize counters and temporary variables
- STEP 9: Loop through the string characters:
- STEP 10: Compare the characters at the start and end pointers
 - If not equal, print "The given string is not a Palindrome" and exit
 - Move the pointers towards the middle of the string
- STEP 11: If the loop completes without printing not a palindrome message:
- STEP 12: Print "The given string is a Palindrome"
- STEP 13: Exit the program

PSEUDO-CODE

Move the address of string message into %rdi register

Call puts function to print the message

Set %rdi to 0

Move the address of the string buffer into %rsi

Set the length of the input buffer to 1024

Set %rax to 0

Perform syscall to read input from user

Move the address of the input string into %rbx

Clear %rcx register

Clear %rdx register

Move the first byte of input string into %cl

Decrement %rax

Decrement %rax

Move the last byte of input string into %dl

Copy %rax to %r14

Decrement %r14

Increment %rax

Set %r15 to 1

Shift right %rax by 1 to find the middle position

Label _find:

Decrement %rax

Compare the characters at the positions stored in %cl and %dl

If they are not equal, jump to _statenot

Move the next character from the input buffer into %cl

Move the previous character from the input buffer into %dl

Increment %r15

Decrement %r14

Compare if we reached the end of the string

If not, jump to _find

Label _stateyes:

Move the address of palindrome message into %rdi register

Call puts function to print the palindrome message

Jump to _end

Label _statenot:

Move the address of not palindrome message into %rdi register

Call puts function to print the not palindrome message

Label _end:

Set %rax to 60 for exit syscall

Clear %rdi register

Perform syscall to exit the program